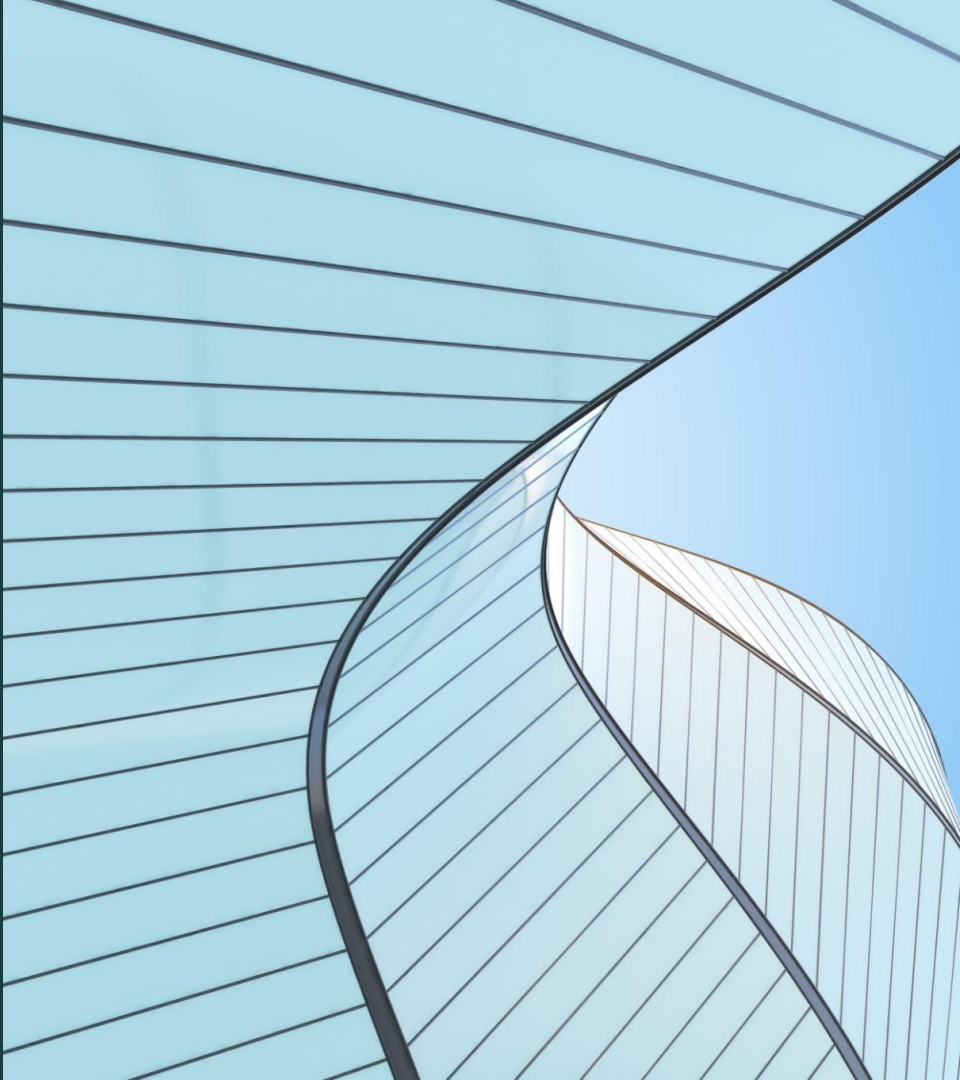


Advanced Retrieval Strategies

*28-day
BOOTCAMP*

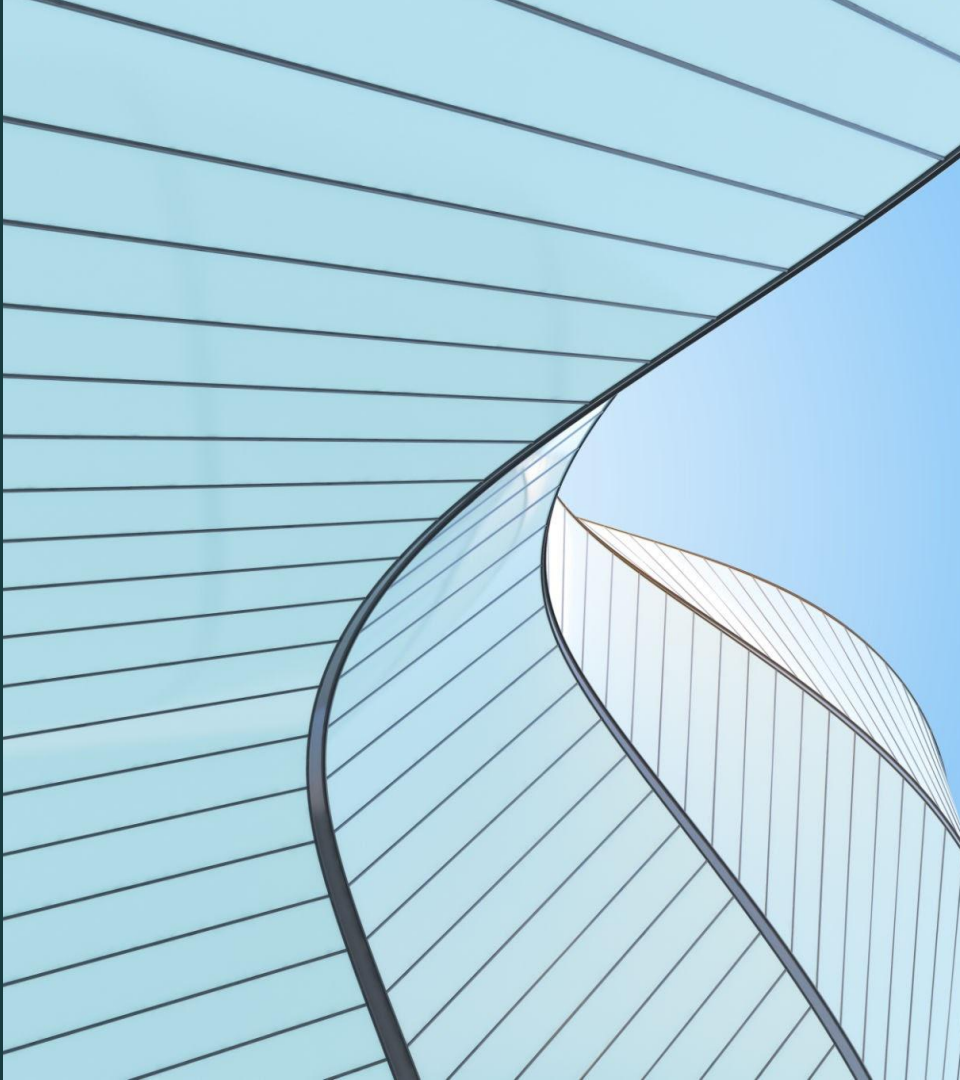
2026

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AGENDA

- Retrieval Efficiency Funnel
- MD Filter
- MMR
- *Advanced Techniques*
 - *Graph-RAG*

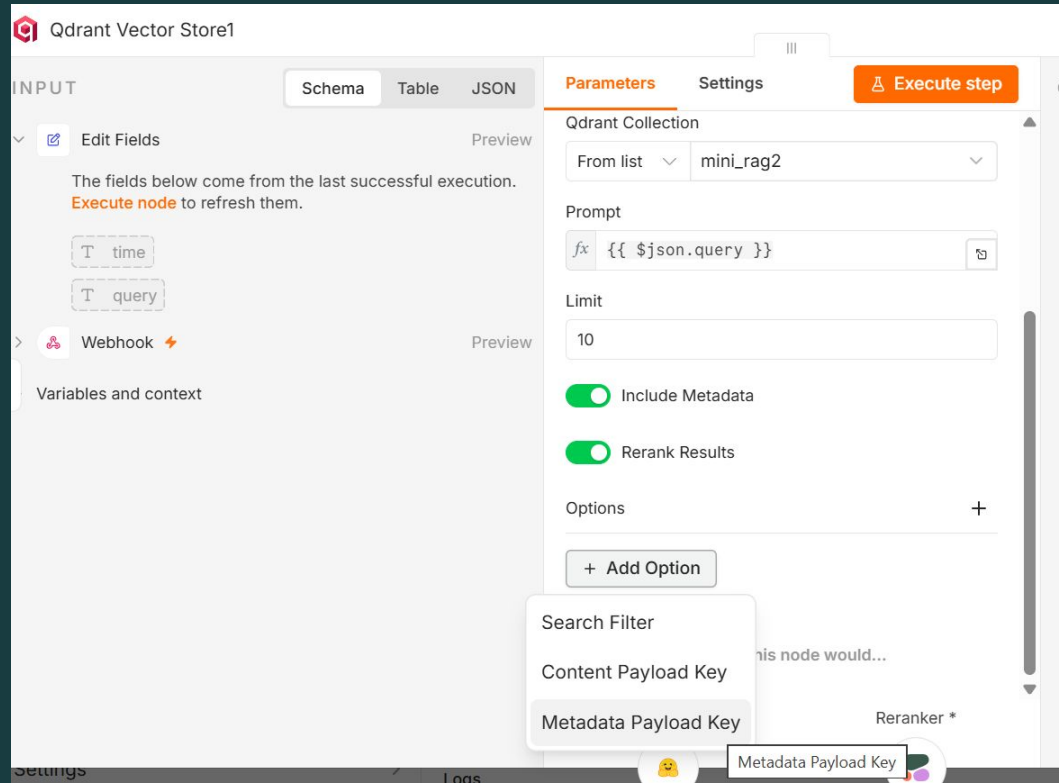


Recap...

- RAG Pipeline
- CP/CR
- Context, Precision trade-off

Metadata Aided Retrieval

- Retrieval Efficiency Funnel
- *Metadata: Rule-based enrichment of chunks*
- MD-filters Improve Precision



Metadata Aided Retrieval

Max Marginal Relevance

$$MMR(d) = \lambda Sim(q, d) - (1 - \lambda) \max_{d' \in S} Sim(d, d')$$

–

Advanced RAG Techniques...

- *Graph-RAG*
 - *Knowledge Graphs*
 - *Multi-hop Reasoning*
- *Multidimensional Relations*
 - *Enrichment of chunks*
- *Scientific Domains*

Summary

- ✓ Metadata based retrieval
 - improves precision
 - saves cost
- ✓ MMR promotes diversity by reducing redundancy
- ✓ Graph-RAG promotes multi-hop reasoning

Assignment, Self-study

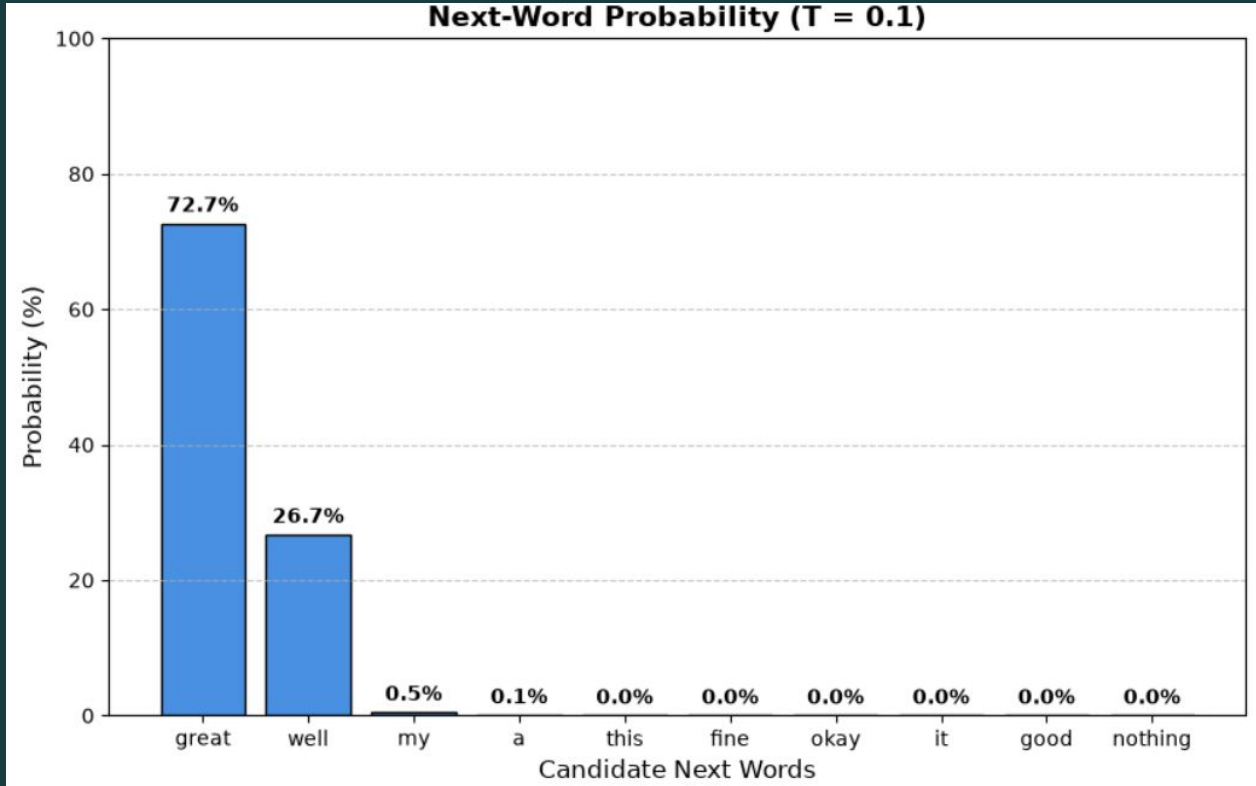
✓ TBD

*Thank
you...*

BACKUP SLIDES

Guess the next, continued...

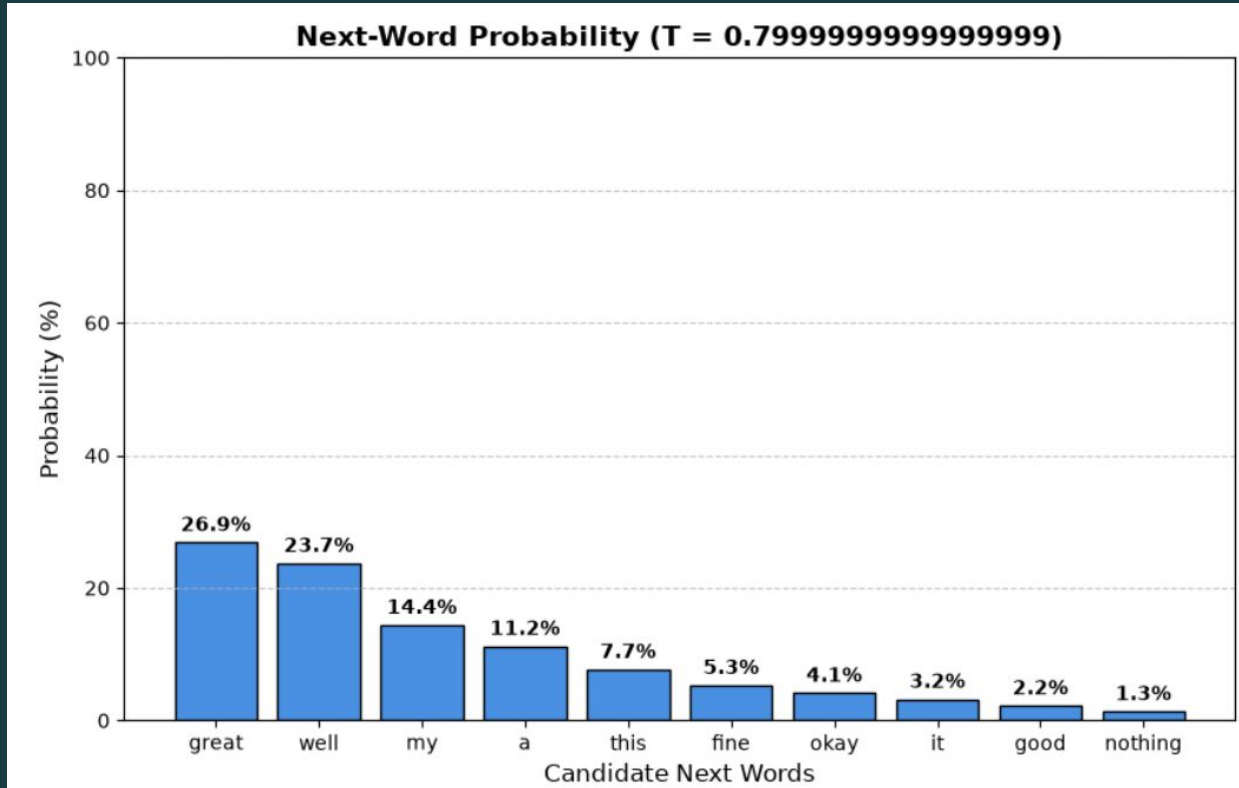
I am doing ?



Temperature = 0.1

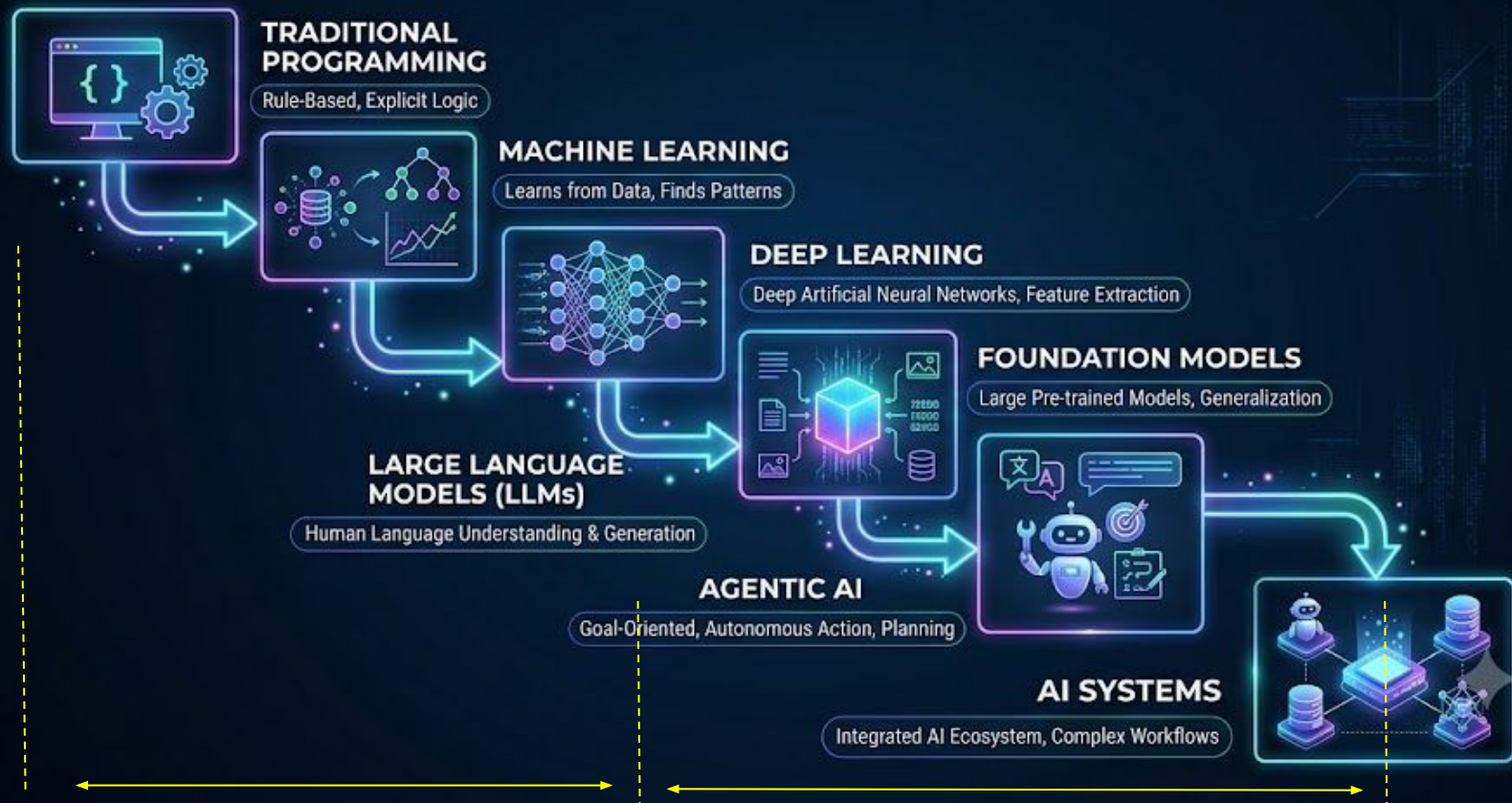
Guess the next, continued...

I am doing ?



Temperature = 0.8

EVOLUTION OF ARTIFICIAL INTELLIGENCE



What are *TOKENS*...?

Why do we care?

- Pricing
- Speed
- Context Window

- Ch-tok
- Word-tok
- Subword
- Byte-Pair-Enc.
- ...

"Artificial Intelligence
is amazing"

['Artificial',
'intelligence', 'is',
'amazing']

"Unbelievable"

['Un', 'believ', 'able']

"Playification"

"Supercalifragilisticexpialidocious"

"こんにちは "

Tokenization

Model	Tokenizer
GPT-2	Byte Pair Encoding (BPE)
GPT-3 / GPT-4 / GPT-5 family	Byte-level BPE (OpenAI variants)
Llama 2 / 3 / 3.2	SentencePiece
Gemma	SentencePiece
Mistral	SentencePiece
Qwen	Byte-level BPE variant
DeepSeek	BPE-based tokenize

Context...

- The number of tokens that for the current input to the LLM – *context*
- The total capacity (in token terms) of input – *context window*
- The model's fundamental unit of computation – *tokens*

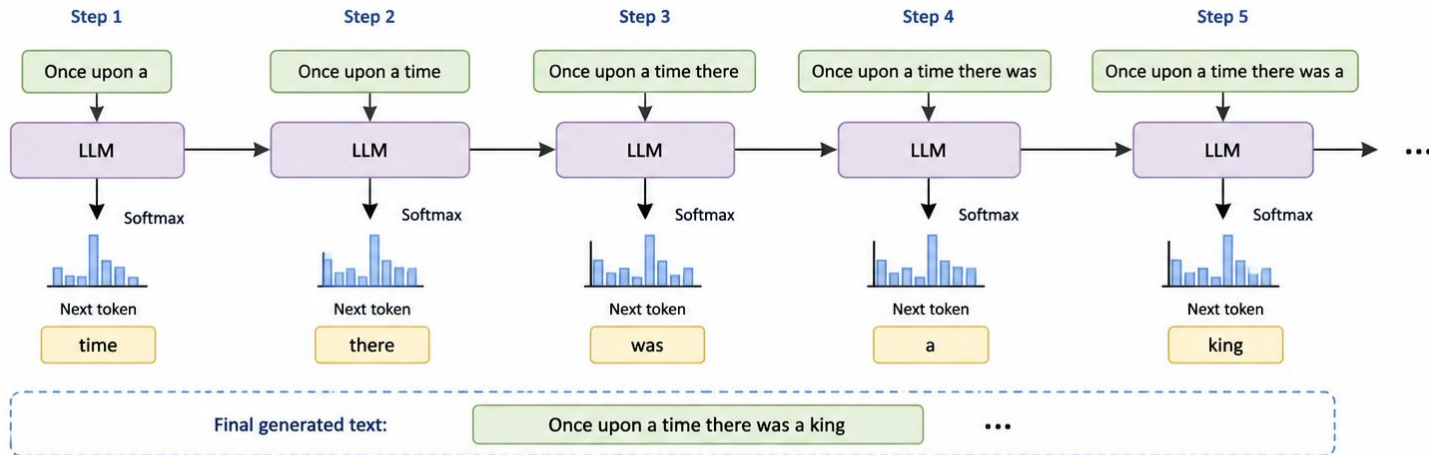
LLM

HIGH LEVEL
ARCH.

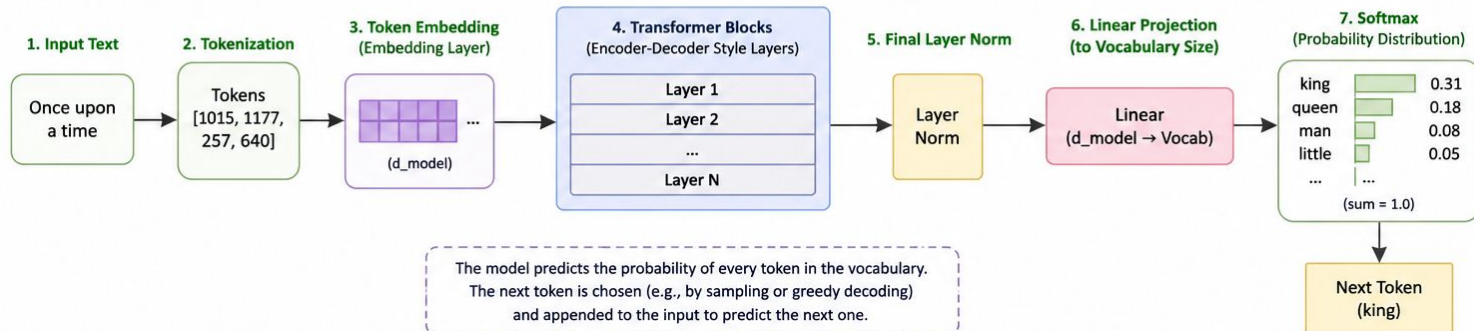
One token
at a time...

AUTOREGRESSIVE GENERATION (NEXT TOKEN PREDICTION)

The model generates one token at a time. Each new token is appended to the context and used to predict the next.



HIGH-LEVEL LLM ARCHITECTURE



WEEK-WISE SCHEDULE

WEEK-1, 2 AI Fundamentals and Evaluation	- LLM fundamentals (evol. From NN / DL) - Prompting architectures - Agentic Foundation (Agentic Arch. patterns) - Metrics: BLEU/ROUGE, Ctx-precision, Recall, Relevance, ... - Intro to RAGAS
WEEK-2, 3 Agentic AI with N8N	- Nodes, Workflows, Triggers, - API Calling, Webhooks, - LLM Integration, Statelessness, Model abstraction - Tool calling, memory (statefulness), Conv. history - Agent Routing, Multi-agent arch.
WEEK-4, 5 RAG Architecture (N8N)	- RAG, EMbedding Fundamentals, - Similarity Search, Vector-DB (Qdrant) - Data-Ingestion, Enrichment, Retrieval Strategies
WEEK- 6 LangFuse, LLMOps	- Telemetry using LiteLLM Proxy - Online and Offline Evals (RAGAS) - LangFuse Dashboarding

INFRA

OPTION 1	OPTION 2
- Local docker-container based setup - Requires Laptop(win) / MacBook - 16GB RAM - 500GB HDD - Installation instructions will be shared for self-hosting. - No guidance provided for setup/installation in the bootcamp	- Hosted on service provide cloud - Use limited-period trial licenses (free period): - N8N - LangFuse - Requires usual laptop - No guidance provided for setup/installation in the bootcamp

AI Licenses

- OpenAI - \$5 token prepaid (optional)
- Groq / Llama3
- Huggingface
- QDrant

LEARNING OUTCOMES

- Understand modern AI architecture
- Build Agentic workflows
- Evaluate AI
- Implement RAG
- Understand LLMOps

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Welcome



Console



Collections



Tutorial



Datasets



Access Tokens



mini_enterprise_rag



AI EMAIL DRAFTING: CONFLICTING EXPECTATIONS & ACCOUNTABILITY

